

1) Landing page

- 1) Add instruction about file format (.csv)
- 2) Add specific column names for each file
 - High-dimensional data ($x_1, x_2, \dots, x_p, ID$)
 - Non-linear dimension reductions ($1-emb1, 1-emb2, 2-emb1, 2-emb2, \dots$)
 - Meta file (Name of the layout, method, hyper-parameter(s))
- 3) Link to the application

Need to give instructions to give name with hyper-parameter underscore /dash with value

I need to find a way to obtain the best default for a_1

2) Data upload

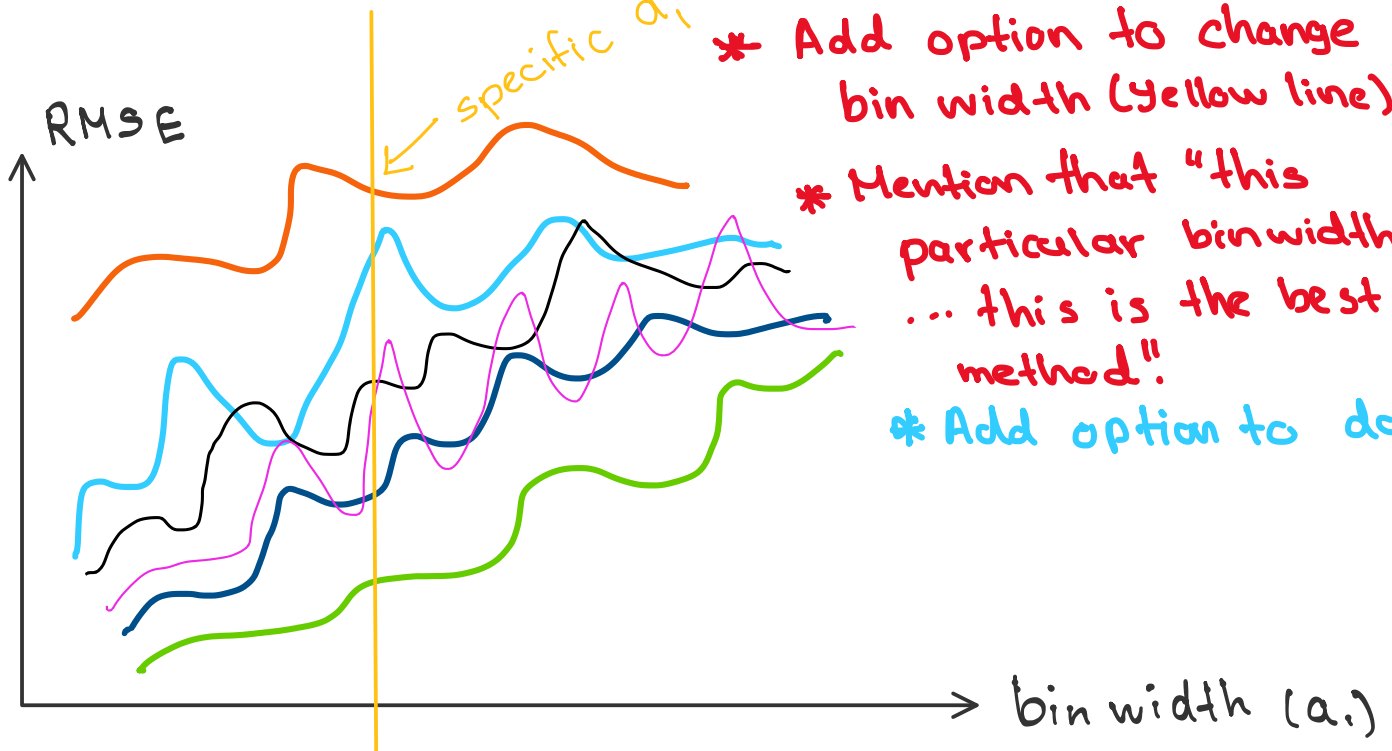
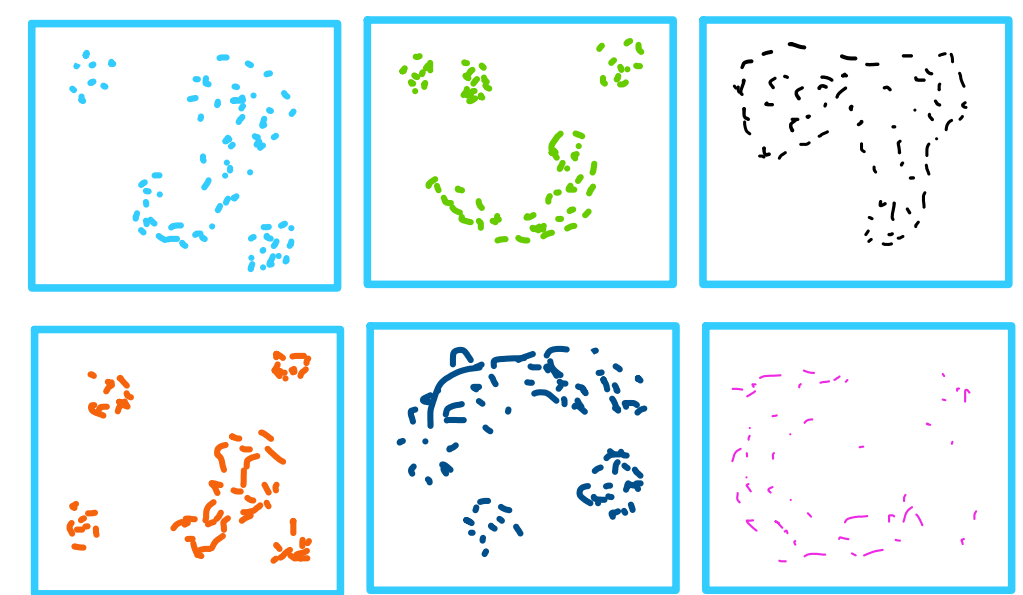
- Upload the high-dimensional data
- Upload the non-linear dimension reduction(s)
- Upload the meta file

* Check file formats and column names before proceed.
* Don't show the tab until all the files uploaded
* check whether we have at least 6 NLDR layouts
* If we have enough, proceed
* If we don't have enough, add UMAP & tSNE layouts (check whether the user uploaded each method, if not add them first, in my opinion)

Why? Having more options will help us to understand the "best" representation more.

- tSNE: perplexity (5, 15, 30, 50 is best)
- UMAP: n-neighbors (5, 15, 30, 50)
- min-dist (0.1 - 0.9)
- still we can ask from users as well.

3) High-d vis model -



- * Add option to change bin width (Yellow line)
- * Mention that "this particular binwidth ... this is the best method!"
- * Add option to download.

Summary of MSE across bin widths (add option to download the summary)

method	a_1	a_2	bin1	bin2	b	b-non-empty	RMSE

- * Note: If there is only one method is uploaded, compute PCA. Because we should have another method to compare with the uploaded one.
- * Add option to download.

- User / analyst obtains "best" representation. (lowest RMSE in particular binwidth)

- Learning how (hyper-) parameter choices affect layout. (Add dropdown with all the methods and ask users to select one, then we can compare the (hyper-)parameters for that particular method and understand what's happening)

4) Diagnostics



* Interactivity with the best layout

By default, take the first or wait until MSE computation and choose the best layout.

- Users can find where the model shows high error and try to understand the reasons.
 - Density of points in the data structure (high error with low densed areas)
 - May be not enough binwidth
- Users can find the mapping b/w 2D layout & high-dimensional data